

does one device (say, a browser) know which computer is the one it is looking for? How does it let the other computer know what it needs? How do technologically different devices communicate with each other? How do they coordinate the exchange seamlessly every time?

Just as the rules functionally define a game, the Internet is functionally defined by a set of protocols, collectively called the Internet Protocol Suite. Today's Internet Protocol Suite began as one of the first research projects for the development of computer networking models, funded by DARPA of the United States Department of Defence. That effort led to the creation of the Transmission Control Protocol/Internet Protocol (TCP/IP) by Robert E. Kahn and Vinton Cerf, which was later split into further layers. TCP/IP provides end-to-end data communication specifying how data should be packetized, addressed, transmitted, routed and received; it tells the Internet how to work, and it is still in use today.

For formulating the TCP/IP, Cerf and Kahn had the intention of enabling the communication of multiple different networks with their own local protocols. They wanted to make a network model that handled the transfer of information as efficiently as possible, with all other intelligence handled at the nodes. Today's Internet Protocol Suite is composed of four layers of abstraction between the information on the computer and the information that is physically transmitted. Within the abstraction layers there are many individual protocols for specific purposes which are as different from each other as sugar and salt. The central two layers are the Internet Layer and Transport Layer, of which the Internet Protocol (IP v4, v6) and Transmission Control Protocol (TCPv4) are major components. The peripheral layers are the Application Layer on the software side, and Link Layer on the hardware side.

What does any of this have to do with websites, you ask? Well, to play a game well, one needs to understand the rules. Similarly, understanding the protocols of the IP Suite will definitely give you a leg-up in your dealings on the Internet. We will discuss the pertinent protocols of the Application Layer and refrain from proceeding down the funnel of abstraction, as they don't pertain specifically to websites.

Application Layer - Protocols for functioning websites

The application layer handles the interface between the message data from host and the message that is communicated. It is not responsible for the communication itself, and has no specifications for how the data is to